

A planar singular minimizer of a strongly polyconvex integral

Jean-Quentin Duquesse
email JeanQuentin.Duquesse@protonmail.com

26/09/2026

Abstract

We describe an explicit Lipschitz map from the disk to $\mathbb{R}^4$, singular only at the origin, which minimizes an autonomous strongly polyconvex integral among all Sobolev competitors with the same trace. The density is nonnegative, has quadratic growth and bounded Hessian, and vanishes precisely on the torus swept out by the gradient. The construction combines an exact supporting inequality with a convex-hull calculation in the space of minors. An affine-envelope extension gives a short proof of the required global density construction.

1 Introduction and statement

For an autonomous variational integral

$$\mathcal{E}_F(v; \Omega) = \int_{\Omega} F(Dv) dx, \quad v : \Omega \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m,$$

we use the convention $Dv \in \mathbb{R}^{m \times n}$, with target components indexing rows. Even smooth strongly convex densities can have singular minimizers when $n \geq 3$. Mooney and Savin [3] constructed a Lipschitz, non- C^1 minimizer in the 2×3 case. Earlier, Šverák and Yan obtained a Lipschitz singular example in 5×3 [5], followed by non-Lipschitz examples in 5×3 and unbounded examples in 14×5 [6]. These examples exhibit increasingly severe failures of regularity as the dimensions increase. In contrast, Morrey's planar regularity theorem [4] excludes non- C^1 minimizers in every $m \times 2$ dimension for smooth strongly convex integrands with bounded Hessian.

The corresponding question for strongly polyconvex integrands requires a distinction between full minimizers and critical points. Bevan [1] constructed planar singular minimizers for strictly polyconvex densities; these densities fail strong ellipticity [2]. Székelyhidi [7] constructed singular Lipschitz critical points, and weak local minimizers, for smooth strongly polyconvex densities in 2×2 . The construction presented here has four target components, quadratic growth and a globally bounded Hessian. To our knowledge, this example is the first of its kind.

For $A \in \mathbb{R}^{4 \times 2}$, let

$$M(A) = (M_{12}, M_{13}, M_{14}, M_{23}, M_{24}, M_{34})(A), \quad J(A) = (A, M(A)) \in \mathbb{R}^{14},$$

where $M_{ij}(A) = A_{i1}A_{j2} - A_{i2}A_{j1}$. We call F *strongly polyconvex with constant $\nu > 0$* if

$$F(A) = \nu|A|^2 + G(J(A)) \tag{1.1}$$

for a convex function G on $\mathbb{R}^{14}$. All matrix norms below are Euclidean.

Theorem 1. *There is a density $F \in C^2(\mathbb{R}^{4 \times 2})$ and constants $c, C > 0$ such that*

$$F \geq 0, \quad c|A|^2 - C \leq F(A) \leq C(1 + |A|^2), \quad \sup_A \|D^2 F(A)\| < \infty, \quad (1.2)$$

$$F(A) - \frac{1}{2}|A|^2 \quad \text{is polyconvex.}$$

Identifying $\mathbb{R}^4$ with $\mathbb{C}^2$, define

$$U(r, \theta) = (re^{i(\theta + \log r)}, re^{2i(\theta + \log r)}), \quad U(0) = 0. \quad (1.3)$$

Then $U \in W^{1,\infty}(B_1; \mathbb{R}^4)$ is smooth on $B_1 \setminus \{0\}$ and is not differentiable at the origin. Moreover,

$$\mathcal{E}_F(U; B_1) = 0 \leq \mathcal{E}_F(v; B_1) \quad \text{for every } v \in U + W_0^{1,2}(B_1; \mathbb{R}^4).$$

The same minimality statement holds on every Lipschitz subdomain of B_1 .

Here “full minimizer” refers to comparison with all the indicated $W^{1,2}$ competitors, without a smallness assumption on the variation or an additional constraint. The proof below in fact permits $F \in C^\infty$; the C^2 statement already includes all the claimed variational properties.

2 The gradient torus and its supports

Let R_t be the counterclockwise planar rotation through angle t , and set

$$A_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \\ 2 & 2 \end{pmatrix}, \quad Q_\phi = \text{diag}(R_\phi, R_{2\phi}), \quad K = \{Q_\phi A_0 R_\psi : (\phi, \psi) \in \mathbb{T}^2\}, \quad (2.1)$$

where $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. Direct differentiation in polar coordinates gives

$$DU(r, \theta) = Q_{\theta + \log r} A_0 R_{-\theta}, \quad |DU|^2 = 12. \quad (2.2)$$

Every element of K occurs at arbitrarily small radii: choose $\theta = -\psi$ modulo 2π and $\log r = \phi + \psi$ modulo 2π . Thus the gradient has the entire torus as its cluster set at the origin.

The parametrization in (2.1) embeds $\mathbb{T}^2$. Indeed, write $B_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ for the first block of A_0 . The identity $R_\phi B_1 R_\psi = B_1$ forces R_ϕ to commute with $B_1 B_1^T$, whose eigenvalues are distinct. Hence $\phi = 0$ or π . The first possibility gives $\psi = 0$; the second gives $\psi = \pi$ and contradicts the second block equation. The stabilizer is therefore trivial, so this compact group orbit is an embedded smooth torus.

Lemma 2. *There is a smooth field $q : K \rightarrow \mathbb{R}^6$ such that*

$$q(A) \cdot M(B - A) \leq -|B - A|^2 \quad (A, B \in K). \quad (2.3)$$

Proof. Let Ω be the alternating form on $\mathbb{R}^4$ with coefficients

$$\omega = (-131, -53, 42, -75, -7, -43) \quad (2.4)$$

in the displayed minor order. For a matrix D with columns D_1, D_2 , write $\Omega(D) = \Omega(D_1, D_2) = \omega \cdot M(D)$. The exact inequality

$$|Q_\phi A_0 R_\psi - A_0|^2 + \Omega(Q_\phi A_0 R_\psi - A_0) \leq 0 \quad (2.5)$$

is proved by the polynomial certificate in Appendix A. At $A = Q_\phi A_0 R_\psi$, define $q(A)$ as the coefficients of

$$\Omega_A(x, y) = \Omega(Q_\phi^T x, Q_\phi^T y).$$

The trivial stabilizer makes this definition smooth and unambiguous. Apply (2.5) to $Q_\phi^T B R_{-\psi} \in K$. Orthogonal transformations preserve the matrix norm, and right multiplication by $R_{-\psi}$ preserves the wedge of the columns. This proves (2.3). $\square$

3 The convex hull of the lifted torus

Lemma 3. *If μ is a probability measure on K and*

$$J(A) = \int_K J(B) d\mu(B) \quad (3.1)$$

for some $A \in \mathbb{R}^{4 \times 2}$, then μ is a point mass. In particular,

$$\text{conv}(J(K)) \cap J(\mathbb{R}^{4 \times 2}) = J(K). \quad (3.2)$$

Proof. Use the unique phase coordinates (ϕ, ψ) on K as random variables with law μ . Right rotations preserve minors. For $d(\phi) = M(Q_\phi A_0)$, direct calculation gives

$$\begin{aligned} d_{12} = 1, \quad d_{34} = 2, \quad (d_{13} - d_{24}) + i(d_{14} + d_{23}) &= ie^{3i\phi}, \\ (d_{13} + d_{24}) + i(d_{23} - d_{14}) &= -3ie^{-i\phi}. \end{aligned} \quad (3.3)$$

Put $m_j = \mathbb{E}e^{ij\phi}$. The Pfaffian of a minor vector is

$$\text{Pf}(d) = d_{12}d_{34} - d_{13}d_{24} + d_{14}d_{23},$$

and (3.3) yields

$$\text{Pf}(\mathbb{E}d) = \frac{8 + |m_3|^2 - 9|m_1|^2}{4}. \quad (3.4)$$

For independent copies ϕ, ϕ' , set $s = \sin((\phi - \phi')/2)$. The identity

$$9|e^{i\phi} - e^{i\phi'}|^2 - |e^{3i\phi} - e^{3i\phi'}|^2 = 32s^4(3 - 2s^2)$$

and $1 - |m_j|^2 = \frac{1}{2}\mathbb{E}|e^{ij\phi} - e^{ij\phi'}|^2$ turn (3.4) into

$$\text{Pf}(\mathbb{E}d) = 4\mathbb{E}[s^4(3 - 2s^2)]. \quad (3.5)$$

Every vector $M(A)$ is decomposable and has Pfaffian zero. Hence (3.1) and (3.5) force ϕ to be constant almost surely.

With ϕ fixed, write $S = \mathbb{E}R_\psi$. Then

$$\mathbb{E}B = Q_\phi A_0 S, \quad M(\mathbb{E}B) = \det(S)d(\phi), \quad \mathbb{E}M(B) = d(\phi).$$

Since $d_{12} = 1$, compatibility requires

$$1 = \det(S) = |\mathbb{E}e^{i\psi}|^2.$$

Equality in this unit-circle averaging inequality forces ψ to be constant almost surely as well. This proves the assertion about μ and, by finite convex combinations, (3.2). $\square$

4 Construction of the density

We now construct a nonnegative density with zero set exactly K . The local extension follows the convex-extension method of Mooney–Savin [3, Lemma 2.1]. Using affine supports instead of supporting paraboloids gives linear growth in the lifted variables and avoids a separate growth-repair argument.

Local convexity in the space of minors

Fix $\nu = 1/2$, extend q smoothly to a tubular neighborhood of K , and prescribe

$$f(A) = a \operatorname{dist}(A, K)^2$$

there. For large constants $a, L > 0$, define

$$H(A, z) = f(A) - \nu|A|^2 + q(A) \cdot (z - M(A)) + L|z - M(A)|^2. \quad (4.1)$$

This function is smooth near $J(K)$. To see that its Hessian is positive there, first differentiate (2.3) along a curve in K . For $T \in T_A K$ this gives

$$q(A) \cdot M(T) \leq -|T|^2.$$

The Hessian of f vanishes on $T_A K$ and equals $2aI$ on its normal complement. Consequently the quadratic form

$$\mathcal{B}_A[V, V] = D^2 f(A)[V, V] - 2\nu|V|^2 - 2q(A) \cdot M(V)$$

is uniformly positive for sufficiently large a : its tangent restriction is at least $|T|^2$, and the normal and mixed terms are controlled by increasing a . Compactness makes this choice uniform in $A \in K$.

At $(A, M(A))$, with $t = \eta - DM(A)[V]$, direct differentiation gives

$$D^2 H[(V, \eta), (V, \eta)] = \mathcal{B}_A[V, V] + 2Dq(A)[V] \cdot t + 2L|t|^2.$$

Choosing L large proves uniform positivity near $J(K)$. Here the term $2q(A) \cdot M(V)$ differentiates the minor map with $q(A)$ held fixed.

The exact quadratic remainder identity

$$M(B) = M(A) + DM(A)[B - A] + M(B - A) \quad (4.2)$$

also gives, for $A, B \in K$,

$$\begin{aligned} H(J(B)) - H(J(A)) - DH(J(A)) \cdot (J(B) - J(A)) &= -\nu|B - A|^2 - q(A) \cdot M(B - A) \\ &\geq \frac{1}{2}|B - A|^2. \end{aligned} \quad (4.3)$$

Since J is Lipschitz on K , this controls $|J(B) - J(A)|^2$. Local Hessian positivity handles nearby pairs; compactness and (4.3) handle separated pairs. Thus there is a compact neighborhood N of $J(K)$, contained in the open region where $D^2 H$ is uniformly positive, and $b > 0$ such that

$$H(X) - H(Y) - DH(Y) \cdot (X - Y) \geq b|X - Y|^2 \quad (X, Y \in N). \quad (4.4)$$

A global convex extension with controlled growth

Define

$$E(X) = \sup_{Y \in N} \{H(Y) + DH(Y) \cdot (X - Y)\}. \quad (4.5)$$

The function E is convex and globally Lipschitz, since N is compact. By (4.4), $E = H$ on N . Let E_ϵ be a smooth mollification and choose $\chi \in C_c^\infty(\text{int } N)$, with $0 \leq \chi \leq 1$, equal to one near $J(K)$. Set

$$G_0 = E_\epsilon + \chi(H - E_\epsilon).$$

On $\text{supp } \chi$, $E_\epsilon \rightarrow H$ in C^2 . Strict positivity of D^2H therefore makes G_0 convex there for small ϵ . Outside this set it equals the convex function E_ϵ . Thus G_0 is globally smooth, convex and Lipschitz, and equals H near $J(K)$. In particular,

$$G_0(J(A)) = -6 \quad (A \in K), \quad |G_0(J(A))| \leq C(1 + |A|^2).$$

We will use a smooth convex maximum $m_\epsilon(s, t)$ which dominates $\max(s, t)$, is nondecreasing in both variables, and agrees with the larger argument when $|s - t| \geq \epsilon$. Explicitly, one may take

$$m_\epsilon(s, t) = \frac{1}{2}(s + t + \rho_\epsilon(s - t)),$$

where ρ_ϵ is the convolution of $|\cdot|$ with a smooth even probability kernel supported in $(-\epsilon, \epsilon)$. For sufficiently small fixed ϵ , put

$$G_1 = m_\epsilon(G_0, -7), \quad F_1(A) = \nu|A|^2 + G_1(J(A)).$$

This leaves $F_1 = f$ near K and gives the global coercive bound

$$F_1(A) \geq \nu|A|^2 - 7. \quad (4.6)$$

Global nonnegativity

Let $S = \text{conv}(J(K))$. If $A \notin K$, Lemma 3 gives $J(A) \notin S$. Strict separation supplies a linear functional s with $s(J(A)) > \max_S s$. For a sufficiently large $N_A > 0$, the affine function

$$\ell_A(X) = N_A(s(X) - \max_S s) - 7 \quad (4.7)$$

satisfies $\nu|A|^2 + \ell_A(J(A)) > 1$, while $\ell_A \leq -7$ on $J(K)$.

Choose a small tubular neighborhood $\mathcal{U}$ of K on which $F_1 = f$. By (4.6), $F_1 > 0$ outside a large ball. The remaining closed set outside $\mathcal{U}$ is compact, so finitely many functions $\ell_1, \dots, \ell_N$ as in (4.7) give strict positivity throughout that set. Take a finite iterated smooth maximum of $G_1, \ell_1, \dots, \ell_N$, denoting the result by G_2 . Each added function has a uniform gap below $G_1 = -6$ on $J(K)$, so the regularization width can be chosen to preserve equality near $J(K)$. The function

$$F_2(A) = \nu|A|^2 + G_2(J(A))$$

is therefore nonnegative, has zero set exactly K , and equals f near K . The lift G_2 remains convex and globally Lipschitz. Hence F_2 has quadratic upper growth, while (4.6) remains valid.

A bounded Hessian at infinity

Define the smooth convex lift

$$\Phi(A, z) = \sqrt{(1 + |A|^2)^2 + |z|^2}, \quad T(A, z) = L\Phi(A, z) - D.$$

Convexity follows by composing the Euclidean norm in (t, z) , which is nondecreasing in $t \geq 0$, with $t = 1 + |A|^2$. Choose L larger than a quadratic upper-growth coefficient for $G_2(J(A))$, and then choose D so large that $T < G_2 - 2$ near $J(K)$. Along the minor graph, $T - G_2 \rightarrow +\infty$ as $|A| \rightarrow \infty$. Thus

$$F(A) = \nu|A|^2 + m_\epsilon(G_2, T)(J(A)), \quad 0 < \epsilon < 1, \quad (4.8)$$

preserves every established property and, outside a large matrix ball, equals

$$\nu|A|^2 + L\sqrt{1 + 2|A|^2 + |A|^4 + |M(A)|^2} - D. \quad (4.9)$$

For the polynomial $P = 1 + 2|A|^2 + |A|^4 + |M(A)|^2$,

$$P \geq (1 + |A|^2)^2, \quad |DP| \leq C|A|(1 + |A|^2), \quad \|D^2P\| \leq C(1 + |A|^2).$$

The identity

$$D^2\sqrt{P} = \frac{D^2P}{2\sqrt{P}} - \frac{DP \otimes DP}{4P^{3/2}}$$

proves the uniform Hessian bound for (4.9). On the remaining compact ball it follows from smoothness. This completes the density construction.

5 Full minimality, singularity and uniqueness

Proof of Theorem 1. The density constructed above has all the properties in (1.2), and its zero set is K . By (2.2), $DU \in K$ away from the origin. Since $|U(x)| = \sqrt{2}|x|$, integration by parts on punctured disks shows that the bounded classical gradient extends as the weak gradient across the origin: the inner boundary term tends to zero. Thus $U \in W^{1,\infty}$.

The first complex component satisfies $U_1(r, 0)/r = e^{i \log r}$, which has no limit as $r \downarrow 0$. Hence U is not differentiable at zero. It is smooth everywhere else. Finally, $F(DU) = 0$ almost everywhere, whereas $F(Dv) \geq 0$ for every competitor. Quadratic upper growth makes every $W^{1,2}$ competitor's energy finite. This proves full minimality on the disk and on its Lipschitz subdomains. $\square$

Proposition 4 (Uniqueness). For the density F constructed above, U is the unique minimizer in $U + W_0^{1,2}(B_1; \mathbb{R}^4)$. The same conclusion holds on every Lipschitz subdomain with boundary trace U .

Proof. Let $V = (v, w) \in \mathbb{C}^2$ be another minimizer. Since $F \geq 0$, $F^{-1}(0) = K$, and the minimum energy is zero, $DV \in K$ almost everywhere. Thus V is Lipschitz and

$$Dv = R_\phi B R_\psi, \quad Dw = R_{2\phi} C R_\psi, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}.$$

In particular, $\det Dv = 1$.

Define the Lipschitz maps

$$P(y) = \frac{y^2}{|y|}, \quad T(y) = ye^{-i \log |y|} \quad (y \neq 0),$$

with $P(0) = T(0) = 0$, and set

$$\Gamma(t) = R_{2t} \operatorname{diag}(1, 2) R_{-t}.$$

Then $DP(y) = \Gamma(\arg y)$ and $Dw = \Gamma(\phi)Dv$. The set $\{v = 0\}$ has measure zero, since $Dv = 0$ almost everywhere on this level set whereas $\det Dv = 1$. Hence, writing $\alpha = \arg v$, the zero-trace map $q = w - P(v)$ satisfies

$$Dq = (\Gamma(\phi) - \Gamma(\alpha))Dv \quad \text{almost everywhere.}$$

A direct calculation gives

$$\det Dq = 2(1 - \cos(\phi - \alpha))^2(2 + \cos(\phi - \alpha)) \geq 0.$$

The null-Lagrangian identity $\int_{B_1} \det Dq = 0$ therefore forces $\phi = \alpha$ modulo 2π almost everywhere. Consequently $Dq = 0$, and the boundary condition gives $w = P(v)$.

Now set $h = T \circ v$. Since $\phi = \alpha$,

$$Dh = R_{\alpha - \log |v|} B^{-1} R_{-\alpha} Dv = R_{\alpha - \log |v|} R_\psi \in SO(2) \quad \text{almost everywhere.}$$

Moreover, $h = x$ on ∂B_1 , so $\int_{B_1} Dh = |B_1|I$. It follows that

$$\int_{B_1} |Dh - I|^2 = 4|B_1| - 2 \int_{B_1} \operatorname{tr} Dh = 0.$$

Thus $h = x$, whence $v(x) = xe^{i \log |x|}$ and $w = P(v)$. Therefore $V = U$.

For a Lipschitz subdomain, extend any competing minimizer by U outside that subdomain and apply the disk result. $\square$

AI disclosure

ChatGPT 6 Astra came up with this construction autonomously.

A An exact support certificate

For $z = \tan(\phi/2)$, the left-hand side of (2.5) is

$$T(z) + V_1(z) \cos \psi + V_2(z) \sin \psi,$$

where

$$\begin{aligned} T(z) &= -\frac{2P_6(z)}{(1+z^2)^3}, \\ V_1(z) &= -\frac{2(51z^4 - 293z^3 - 15z^2 + 159z - 46)}{(1+z^2)^2}, \\ V_2(z) &= -\frac{2z(72z^3 + 309z^2 - 216z + 61)}{(1+z^2)^2}, \\ P_6(z) &= 205z^6 - 159z^5 + 588z^4 + 50z^3 + 165z^2 - 159z + 46. \end{aligned}$$

Direct expansion gives

$$T(z)^2 - V_1(z)^2 - V_2(z)^2 = \frac{4z^2 P_{10}(z)}{(1+z^2)^6},$$

with

$$P_{10}(z) = 34240z^{10} - 79800z^9 + 102095z^8 - 96008z^7 + 105604z^6 \\ + 130088z^5 + 113458z^4 - 62456z^3 + 15828z^2 - 14448z + 5847.$$

For the exact rational Sturm recurrence $p_0 = P$, $p_1 = P'$, $p_{j+1} = -\text{rem}(p_{j-1}, p_j)$, the endpoint signs are

Polynomial	Signs at $-\infty$	Signs at $+\infty$	Variations
P_6	$+, -, -, +, +, -, -$	$+, +, -, -, +, +, -$	3, 3
P_{10}	$+, -, -, -, +, -, -, +, -, -, -$	$+, +, -, +, +, +, -, -, -, +, -$	5, 5

Neither polynomial has a real root, and $P_6(0) = 46$, $P_{10}(0) = 5847$. Thus both are positive, $T < 0$, and $T^2 \geq V_1^2 + V_2^2$, proving (2.5). The omitted phase $\phi = \pi$ gives $(T, V_1, V_2) = (-410, -102, -144)$ and a strict inequality. At $z = 0$ the expression is $-92 + 92 \cos \psi$, so equality occurs only at the anchor.

References

- [1] J. J. Bevan, *Singular minimizers of strictly polyconvex functionals in $\mathbb{R}^{2 \times 2}$* , Calc. Var. Partial Differential Equations **23** (2005), 347–372. doi:10.1007/s00526-004-0305-6.
- [2] J. J. Bevan, *On one-homogeneous solutions to elliptic systems with spatial variable dependence in two dimensions*, Proc. Roy. Soc. Edinburgh Sect. A **140** (2010), 449–475. doi:10.1017/S0308210508000516.
- [3] C. Mooney and O. Savin, *Some singular minimizers in low dimensions in the calculus of variations*, Arch. Ration. Mech. Anal. **221** (2016), 1–22. doi:10.1007/s00205-015-0955-x.
- [4] C. B. Morrey, Jr., *Multiple Integrals in the Calculus of Variations*, Grundlehren der mathematischen Wissenschaften, vol. 130, Springer, 1966. doi:10.1007/978-3-540-69952-1.
- [5] V. Šverák and X. Yan, *A singular minimizer of a smooth strongly convex functional in three dimensions*, Calc. Var. Partial Differential Equations **10** (2000), 213–221. doi:10.1007/s005260050151.
- [6] V. Šverák and X. Yan, *Non-Lipschitz minimizers of smooth uniformly convex functionals*, Proc. Natl. Acad. Sci. USA **99** (2002), 15269–15276. doi:10.1073/pnas.222494699.
- [7] L. Székelyhidi, Jr., *The regularity of critical points of polyconvex functionals*, Arch. Ration. Mech. Anal. **172** (2004), 133–152. doi:10.1007/s00205-003-0300-7.